

2.3.1 Limits Worksheet

1. Find the following limits (if they exist). If the limit does not exist explain why.

a. Find $\lim_{t \rightarrow -1} f(t)$

$$\lim_{t \rightarrow -1} f(t) = 1$$

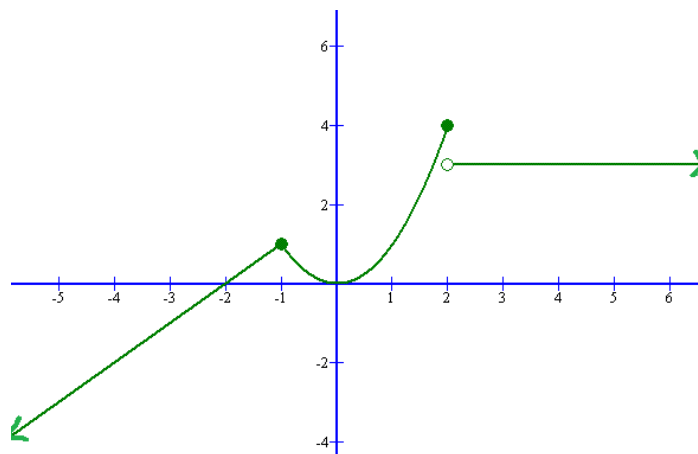
b. Find $\lim_{t \rightarrow 2^+} f(t)$

$$\lim_{t \rightarrow 2^+} f(t) = 3$$

c. Find $\lim_{t \rightarrow 2} f(t)$

$$\lim_{t \rightarrow 2} f(t) \text{ does not exist.}$$

$$\text{This is because } \lim_{t \rightarrow 2^+} f(t) \neq \lim_{t \rightarrow 2^-} f(t)$$



Problems #2-12: Evaluate the limit, if it exists. If you claim that it does not exist (DNE) then provide a valid argument to support your claim.

$$2. \lim_{x \rightarrow -4} \frac{x^2 + 5x + 4}{x^2 + 3x - 4} = \lim_{x \rightarrow -4} \frac{(x+4)(x+1)}{(x+4)(x-1)} = \lim_{x \rightarrow -4} \frac{(x+1)}{(x-1)} = \frac{-4+1}{-4-1} = \frac{-3}{-5} = \frac{3}{5}$$

$$3. \lim_{x \rightarrow 4} \frac{x^2 - 4x}{x^2 - 3x - 4} = \lim_{x \rightarrow 4} \frac{x(x-4)}{(x-4)(x+1)} = \lim_{x \rightarrow 4} \frac{x}{(x+1)} = \frac{4}{4+1} = \frac{4}{5}$$

$$4. \lim_{x \rightarrow -1} \frac{x^2 - 4x}{x^2 - 3x - 4} = \lim_{x \rightarrow -1} \frac{x(x-4)}{(x-4)(x+1)} = \lim_{x \rightarrow -1} \frac{x}{(x+1)}, \text{ we begin to expect the limit DNE.}$$

Conclusion

$$\lim_{x \rightarrow -1^-} \frac{x}{(x+1)} = +\infty$$

$$\lim_{x \rightarrow -1^+} \frac{x}{(x+1)} = -\infty$$

Thought Process

$$\lim_{x \rightarrow -1^-} \frac{x}{(x+1)} \rightarrow \frac{-1}{0^-} \rightarrow +\infty$$

$$\lim_{x \rightarrow -1^+} \frac{x}{(x+1)} \rightarrow \frac{-1}{0^+} \rightarrow -\infty$$

Therefore, $\lim_{x \rightarrow -1} \frac{x}{(x+1)}$ does not exist.

$$5. \lim_{h \rightarrow 0} \frac{\sqrt{1+h}-1}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{1+h}-1}{h} \cdot \frac{\sqrt{1+h}+1}{\sqrt{1+h}+1} = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{1+h}+1)} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{1+h}+1} = \frac{1}{\sqrt{1+0}+1} = \frac{1}{2}$$

$$6. \lim_{t \rightarrow 0} \left(\frac{1}{t} - \frac{1}{t^2+t} \right) = \lim_{t \rightarrow 0} \left(\frac{(t+1)}{t(t+1)} - \frac{1}{t(t+1)} \right) = \lim_{t \rightarrow 0} \frac{t}{t(t+1)} = \lim_{t \rightarrow 0} \frac{1}{(t+1)} = \frac{1}{0+1} = 1$$

$$7. \lim_{x \rightarrow 2.5^+} \frac{e^x}{5-2x} \rightarrow \frac{e^{2.5^+}}{0^-} \rightarrow -\infty \quad \text{Therefore, } \lim_{x \rightarrow 2.5^+} \frac{e^x}{5-2x} = -\infty$$

$$8. \lim_{x \rightarrow 16} \frac{4-\sqrt{x}}{16x-x^2} = \lim_{x \rightarrow 16} \frac{4-\sqrt{x}}{x(16-x)} = \lim_{x \rightarrow 16} \frac{4-\sqrt{x}}{x(4-\sqrt{x})(4+\sqrt{x})} = \lim_{x \rightarrow 16} \frac{1}{x(4+\sqrt{x})} = \frac{1}{16(4+\sqrt{16})} = \frac{1}{128}$$

This problem could also be done by multiplying by the conjugate $\left(\frac{4+\sqrt{x}}{4+\sqrt{x}} \right)$.

$$9. \lim_{x \rightarrow -1} \frac{x^2+2x+1}{x^4-1} = \lim_{x \rightarrow -1} \frac{(x+1)^2}{(x^2-1)(x^2+1)} = \lim_{x \rightarrow -1} \frac{(x+1)^2}{(x+1)(x-1)(x^2+1)} = \lim_{x \rightarrow -1} \frac{(x+1)}{(x-1)(x^2+1)} = \frac{-1+1}{(-1-1)((-1)^2+1)} = 0$$

$$10. \lim_{x \rightarrow -2} \frac{2-|x|}{2+x} \quad (\text{Hint: If } x \rightarrow -2 \text{ what can you say about the value of } |x| \text{ ?})$$

If $x < 0$ then $|x| = -x$. We can use this to make a substitution.

$$\lim_{x \rightarrow -2} \frac{2-|x|}{2+x} = \lim_{x \rightarrow -2} \frac{2-(-x)}{2+x} = \lim_{x \rightarrow -2} \frac{2+x}{2+x} = \lim_{x \rightarrow -2} 1 = 1$$

$$11. \text{ Consider the function } f(x) = \begin{cases} 1 + \sin(x) & \text{if } x < 0 \\ \cos(x) & \text{if } 0 \leq x \leq \pi \\ \sin(x) & \text{if } x > \pi \end{cases}$$

a) Find $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$.

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} [1 + \sin(x)] = 1 + \sin(0) = 1 + 0 = 1$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \cos(x) = \cos(0) = 1$$

b) Does $\lim_{x \rightarrow 0} f(x)$ exist?

Yes, the limit exists.

Since $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = 1$ we know that $\lim_{x \rightarrow 0} f(x) = 1$.

c) Find $\lim_{x \rightarrow \pi^-} f(x)$ and $\lim_{x \rightarrow \pi^+} f(x)$.

$$\lim_{x \rightarrow \pi^-} f(x) = \lim_{x \rightarrow \pi^-} \cos(x) = \cos(\pi) = -1$$

$$\lim_{x \rightarrow \pi^+} f(x) = \lim_{x \rightarrow \pi^+} \sin(x) = \sin(\pi) = 0$$

d) Does $\lim_{x \rightarrow \pi} f(x)$ exist?

No, the limit does not exist.

Since $\lim_{x \rightarrow \pi^-} f(x) \neq \lim_{x \rightarrow \pi^+} f(x)$ we know that $\lim_{x \rightarrow \pi} f(x)$ *does not exist*.

12. Consider the function $g(x) = \begin{cases} x^2 & \text{if } x < 1 \\ \sqrt{x} & \text{if } x \geq 1 \end{cases}$.

a) Find $\lim_{x \rightarrow 0} g(x)$.

We can check this limit directly since $x = 0$ is not a potential break point of the graph.

$$\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} x^2 = (0)^2 = 0$$

b) Find $\lim_{x \rightarrow 1} g(x)$.

We should check the limit from both sides, since a potential break occurs at $x = 1$. We see that

$$\lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^-} x^2 = (-1)^2 = 1 \quad \& \quad \lim_{x \rightarrow 1^+} g(x) = \lim_{x \rightarrow 1^+} \sqrt{x} = \sqrt{1} = 1$$

Therefore, since $\lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^+} g(x) = 1$ we know that $\lim_{x \rightarrow 1} g(x) = 1$.

13. Use the Squeeze Theorem to help determine the following limit: $\lim_{x \rightarrow 0} x^3 \cos\left(\frac{1}{x^2}\right)$.

By properties of the cosine function, we know that $-1 \leq \cos\left(\frac{1}{x^2}\right) \leq 1$.

Therefore, $-x^3 \leq x^3 \cos\left(\frac{1}{x^2}\right) \leq x^3$.

Since $\lim_{x \rightarrow 0} -x^3 = 0$ and $\lim_{x \rightarrow 0} x^3 = 0$, then by the Squeeze Theorem $\lim_{x \rightarrow 0} x^3 \cos\left(\frac{1}{x^2}\right) = 0$.

14. Use the Squeeze Theorem to help determine the following limit: $\lim_{x \rightarrow -\infty} \frac{5x^2 - \sin(3x)}{x^2 + 10}$.

By properties of the sine function, we know that $-1 \leq \sin(3x) \leq 1$.

Therefore, $-1 \leq -\sin(3x) \leq 1$.

Therefore, $5x^2 - 1 \leq 5x^2 - \sin(3x) \leq 5x^2 + 1$.

Therefore, $\frac{5x^2 - 1}{x^2 + 10} \leq \frac{5x^2 - \sin(3x)}{x^2 + 10} \leq \frac{5x^2 + 1}{x^2 + 10}$.

Since $\lim_{x \rightarrow -\infty} \frac{5x^2 - 1}{x^2 + 10} = \lim_{x \rightarrow -\infty} \frac{5 - \frac{1}{x^2}}{1 + \frac{10}{x^2}} = \frac{5 + 0}{1 + 0} = 5$ and $\lim_{x \rightarrow -\infty} \frac{5x^2 + 1}{x^2 + 10} = \lim_{x \rightarrow -\infty} \frac{5 + \frac{1}{x^2}}{1 + \frac{10}{x^2}} = \frac{5 + 0}{1 + 0} = 5$, then by the

Squeeze Theorem $\lim_{x \rightarrow -\infty} \frac{5x^2 - \sin(3x)}{x^2 + 10} = 5$.